1. What is AWS?
AWS stands for Amazon Web services. It is a cloud platform that provides services like computing, storage, databases, networking, security and monitoring over the internet.

2. Cloud computing?
Cloud computing means using computing resources such as servers, storage and databases over the internet instead of maintaining physical servers ourselves.

3. Advantages of CC -> * No need to buy physical servers, Easy to scale
2. pay only for what we use,
3. High availability
4. Faster deployment
5. Global access.

4. Cloud service models?
IAAS - Infrastructure as a service
PAAS - Platform as a service
SAAS - Software as a service.

5. 1. IAAS -> provides infrastructure such as virtual machines, networking and storage.
Example : Amazon EC2
2. PAAS -> PAas provides a platform where developers can build and deploy applications without managing the underlying infrastructure.
3. SAAS - Saas provides ready-to-use software through the internet.
Example : Gmail or Microsoft 365.

2. AWS GLOBAL INFRASTRUCTURE

8. What is an AWS Region?

A region is a geographical area where AWS has multiple data centers.

Example: Mumbai is an AWS Region in India.

9. what is an Availability Zone?

An Availability Zone, or AZ, is one or more isolated data centers within an AWS Region.

10. Diff Region & Availability Zone

Region	Availability Zone
1. Geographical location	Isolated location inside a region
2. contains multiple AZs	Belongs to one region
Example: Mumbai	Example: an AZ within Mumbai

11. What is fault tolerance?

Fault tolerance means a system can continue working even if one component fails.

3. AWS Management Console:

14. AWS Management Console?

It is a web-based graphical interface used to create, configure and manage AWS resources.

15. How can you interact with AWS

1. AWS Management Console
2. AWS CLI → allows us to manage AWS resources using commands
3. AWS SDK

uses → Automation
scripting
Faster resource management
Repeated tasks

18. AWS Cloudshell → is a browser-based command-line environment provided by AWS. We can run AWS CLI commands without installing the CLI locally.

3. CLI vs console?
Console: Graphical interface
CLI: command-line Interface
For example, instead of creating an S_3 bucket through the console, we can use an AWS CLI command.

4. IAM:
↳ helps us to provide secure & controlled access to AWS resources.
Main components of IAM → Users, Groups, Roles, Polic

23. IAM User:
An IAM user represents a person or application that needs access to AWS resources.

24. IAM Group:
A group is a collection of IAM users.
For example, we can create a Developers group and give it permissions required by developers.

25. IAM Policy:
A policy is a document that defines what actions are allowed or denied on AWS resources.

26. What is an IAM Role?
It provides temporary permissions to a user, application or AWS service accou without requiring permanent access keys.

27. User vs Role
↓ represents specific identity
↳ Provides permissions that can be assumed temporarily by an identity or service.

Least privilege → Give a user or application only the permissions it needs, and nothing extra. Reduces risk of unauthorized access and limits the damage if credentials are compromised.

S3

S3 stands for Simple Storage Service. It is an object storage service used to store files such as images, videos, documents and backups.

S3 Bucket → is a container used to store objects in S3.

S3 object → is the actual file and its associated metadata stored in an S3 bucket.

Bucket → Container

object → File stored inside the bucket.